

Data Structure Design

0.x

CartoObject

Dictionary<string, List<float3>> Edges

Storing verticies in a list, and then in a dictionary to distinguish different geometries. Supported geometries are: *CenterLine*, *Edge* & *Location*.

Dictionary<string, object> Properties

Storing properties in a dictionary. Supported properties are: *Address*, *Area*, *Asset*, *Brand*, *Category*, *Center*, *Color*, *Company*, *Density*, *Direction*, *Employee*, *Form*, *Height*, *Household*, *Length*, *Level*, *Name*, *Object*, *Product*, *Resident*, *Theme*, *Unlocked*, *Width* & *Zoning*.

Dictionary<string, string> Type

Storing spatial type in a dictionary. Supported properties are: *None*, *Point*, *LineString*, *Polygon*, *Raster*.

Dictionary<string, float[,]> Values

Storing raster in a dictionary. Supported values are: *Depth*, *Elevation*, *WorldDepth*, *WorldElevation*.

Geodata

List<CartoObject> m_Objects

Storing features in a list.

Dictionary<string, int> FieldLength

Storing the maximum length for each field; used for Shapefiles.

Exportable Objects

Building: *property boundary*
District: *boundary*
Map tile: *boundary*
Path: *boundary*, *centerline*
Road: *centerline*, *boundary*
Track: *centerline*, *boundary*
Zoning: *cell*

POI: *location*

Terrain: (world) *terrain*
Water bodies: (world) *water bodies*

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Feature

Dictionary<VectorKind, Geometry> Geometries

Defines the geometry of each category. Valid keys are: *CenterLine*, *Boundary*, *Footprint* & *Location*.

Dictionary<Property, object> Properties

Defines the properties of the feature. Valid keys are: *Address*, *Asset*, *Brand*, *Capacity*¹, *Category*, *Color*, *Company*, *Density*, *Direction*, *Discharge*², *Elevation*³, *Employee*, *Form*, *Height*⁴, *Household*, *Length*, *Level*, *Limit*⁵, *Load*⁶, *Model*⁷, *Name*, *Object*, *Passenger*⁸, *Product*, *Resident*, *Story*⁹, *Stop*¹⁰, *Theme*, *Transport*¹¹, *Unlocked*, *Value*¹², *Vehicle*¹³, *Volume*¹⁴, *Wealth*¹⁵, *Width* & *Zoning*.

Geodata

Geometry

float3[][] Exclusions

Defines the region that removes area from a polygon.

float3[][] Inclusions

Defines the region that forms area to the polygon.

RasterData : Geodata

Dictionary<RasterKind, float[][]> Grids

Defines the grid of each category. Valid keys are: *Direction*¹⁶, *Deposit*¹⁷, *Depth*, *Elevation*, *Intensity*¹⁸, *WorldDepth* & *WorldElevation*.

VectorData : Geodata

List<Feature> Features

Defines all the features.

Dictionary<string, int> Lengths

The maximum length of each property. See *Feature.Properties* for all valid keys.

Exportable Objects

Building: *footprint*, *property boundary*
District: *boundary*
Extractor & Landfill: *footprint*
Map tile: *boundary*
Path: *boundary*, *centerline*
Road: *centerline*, *boundary*
Route: *centerline*
Surface: *boundary*
Track: *centerline*, *boundary*
Utility: *centerline*
Zoning: *cell*

POI: *location*

Land Value: *intensity*
Pollution: *intensity*
Resources: *deposit*
Terrain: (world) *elevation*
Water bodies: (world) *depth*, *direction*, *intensity*
Wind: *direction*, *intensity*

Side notes

- 1 The capacity of the pipe.
- 2 The volumetric flow rate of the fluids.
- 3 The equivalent of 0.x's *Height* field.
- 4 The building's height.
- 5 The speed liit of the network.
- 6 The amount of the electricity.
- 7 The title of the model operating on the route.
- 8 The number of passengers.
- 9 The number of levels of the building.
- 10 The number of stops on the route.
- 11 The transportation operating on the route.
- 12 The land value of the building.
- 13 The number of vehicles operating on the route.
- 14 The average volume of the road.
- 15 The average wealth of the residents.
- 16 The direction of the flow or wind.
- 17 The deposit of natural resources.
- 18 The intensity of pollution or land value.

Modification information (since v25-01-05)

- 1 *Geodata.Shapes (Removed)*
 - > Can be implemented by a dictionary, since the *VectorKind*-*Shape* relationship is consistent.
- 2 *Feature.Geometries (Modified)*
Feature.Properties (Modified)
RasterData.Grids (Modified)
 - > The key of these dictionaries had been replaced by corresponding enumerations for consistency.

Export Process Design

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Utils.ExportUtils.ExportToFile()

Triggered when user clicks *Export* button.

- 1 Log user export options.
- 2 Validate export options:
 - 2-1 Validate latitude.
 - 2-2 Validate longitude.
 - 2-3 Validate export file accessibility.
 - 2-4 Validate spatial field integrity.
- 3 Raise error dialog if any validation fails.
- 4 Create related folders if they don't exist.
- 5 Export rasters if the file is in raster format:
 - 5-1 Validate the feature checkboxes.
 - 5-2 *Systems.XxSystem.GetXxProperties()*
 - 5-3 Turn *List<CartoObject>* into *Geodata*.
 - 5-4 *ExportRasterGeodata()*
- 6 Export vectors if the file is in vector format:
 - 6-1 Validate the feature checkboxes.
 - 6-2 *Systems.XxSystem.GetXxProperties()*
 - 6-3 Turn *List<CartoObject>* into *Geodata*.
 - 6-4 *ExportVectorGeodata()*
- 7 Raise the message dialog when finishes.

Systems.XxSystem.GetXxProperties()

Query and gather essential geometries, properties, and grids.

- 1 Initialization (e.g. querying pre-requirements).
- 2 For every instance in the query result:
 - 2-1 Validate the field checkboxes.
 - 2-2 Fill the properties and geometries.
 - 2-3 Fill the grid.
 - 2-4 Combine the result into a *CartoObject*.
 - 2-5 Add it to *List<CartoObject>*.
- 3 Return *List<CartoObject>*.

ExportRasterGeodata()

ExportVectorGeodata()

Both are the wrappers of *ExportToFormat()* .

ExportToFormat()

Trigger input geodata's *Geodata.ToXx()* .

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IO.Export()

Triggered when user clicks *Export* button.

- 1 Log user export options.
- 2 Validate export options:
 - 2-1 *Not decided yet...*
- 3 Raise error dialog if any validation fails.
- 4 Create related folders if they don't exist.
- 5 Export files in searched order. *Need futher considerations on the design.*